INDICATION CALCULATOR · CONTROLLING & FINANCE · PRE-CONFERENCE DATEN-WG 2026

Visual standards calculator: Paid, Flexible or MacGyver?

What does it cost to implement visual standards in Power BI with **3rd-party visuals**, **open source**, **Deneb** or **core visuals**? The guiding principle: **standard first, only as bespoke as necessary.** So the calculator first checks whether there is a professional reason to depart from the core visuals, and then compares across three levels: Overview with three inputs, Narrow down via requirements and red flags, Details with every assumption and source. It gives an **indication** with ranges, not a price quote: every assumption is editable and marked with its evidence. **If this is your first time here, answer seven short questions in the guided start** — it only sets fields that you will find again, every one of them, in levels 2 and 3.

LEVEL 1 What does a standard cost you, per viewer per year?

Three inputs, everything else sits at its default: a five-year horizon, volume tiering (the more licences, the cheaper each one) and a mid-range chart mix. In each approach's bar the band shows the P10 to P90 range, and the line inside it shows **P50 — the middle number: in half of the simulation runs the cost falls below it**. Levels 2 and 3 get more precise.

Presets: Pilot · 1 report author / 10 viewers | Mid-market · 3 / 40 | Corporate · 25 / 1,500 | Corporate with Report Server · 15 / 400 | Large corporate · 200 / 10,000

Scenarios: Enterprise with Org Store · 25 / 1,500 | Public administration · 6 / 300 | **Finance with mandatory IBCS · 5 / 150** | Replacing an installed base · 8 / 400 | Build instead of buy · 1 / 10

Finance with mandatory IBCS · 5 / 150: The finance department the standard is made for: mostly management reporting, every report to the standard. So with Paid every reader needs a licence, and class A disappears. Not suited if only part of the reports need the standard — adjust the report classes instead.

Report authors: 5

People who build reports with the standard visuals.

Viewers (read-only): 150

With paid visuals, only the share of viewers who open class B or C reports needs a licence (default 85%, slider in level 3; with mandatory IBCS, all of them).

Operating model

Enterprise BI | **Managed Self-Service** | Self-Service

A core team maintains templates, trained report authors use them.

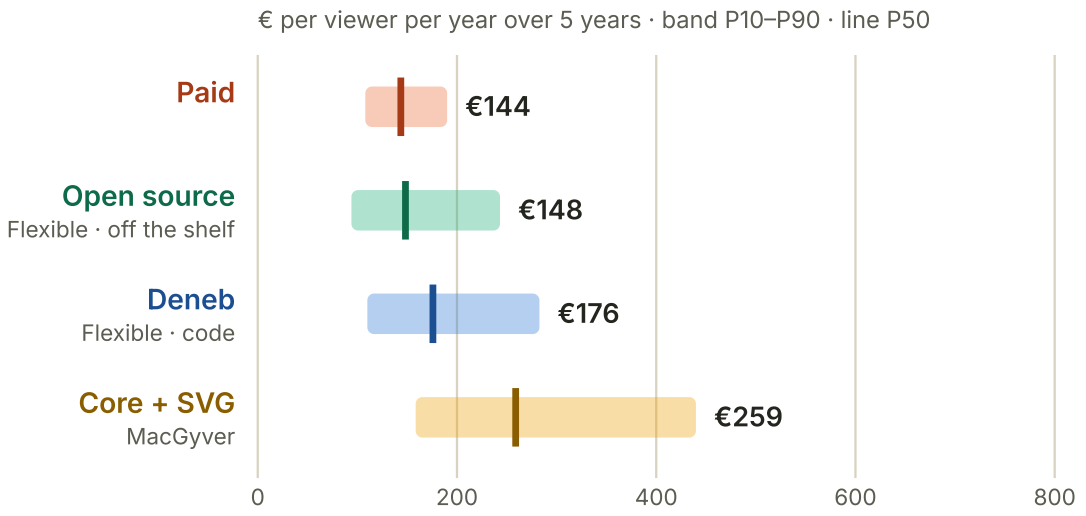
Price anchor for paid

Inforiver ~€33 | **Blended price ~€80** | Zebra level ~€170 (derived)

Per **licensed** user per year before volume tiering (licensed headcount: all report authors plus the share of viewers with access to standard reports). It flips the ranking more than any other assumption; the sentence on the right shows all three.

☐ A ready-made template library for Deneb and core is available
e.g. IBCS Chart Builder (the Daten-WG's Deneb templates, our own project), vegaviz, PowerofBI UDFs for SVG. Cuts initial build and reuse to 60 to 70%, training by half as much. Off by default, so that our own project does not quietly count in its own favour.

With 5 report authors and 150 viewers, **Paid and Open source** are level at around €144 per viewer per year (P50); Deneb costs 1.2 times as much, Core + SVG costs 1.8 times as much. From around **176 viewers** the licence per head becomes more expensive than the cheapest hours-based approach. Price anchor: Blended price €80 · without a template library · hours anchor: Panel (default). **13%** of the Paid cost rests on evidenced prices, **43%** on estimates (the rest: rules of thumb and the price anchor). The bar per approach is in level 2.



WHY THIS RESULT, AND WHAT TURNS IT

1 · Ratio 1 : 30. Paid pays €58,000 in licences over 5 years. Open source saves that, but needs 528 more hours, valued at €66,000. The two weigh about the same here, so the result is a tie.

2 · Licence price. Anchor €80 per licensed user per year (list price before tiering). Parity with Open source at around **€92**. That is the negotiation ceiling, not the list price.

3 · Hours per chart. Panel values per reused chart: paid 1 h, open source 2 h, Deneb 2.5 h, Core 4 h (unmeasured). Open source would have to cut its hours by around **6%** to draw level; the switch "template library" cuts Deneb and Core by 30 to 40%.

What holds is the direction of these three levers, not the second digit. The number of viewers, the licence price and the hours per chart explain the gap almost completely; everything else shifts it by percentage points. Key statements for discussion

What holds: the order of magnitude per viewer and the number of viewers from which the licence dominates (structural). **What does not hold:** the order of Paid and Open source, which hangs on unmeasured hour assumptions; amounts to more than one digit, and the benefit side. Provenance of the 58 assumptions: 3 verified, 9 search snippets, 3 rules of thumb, 9 simulated panel, 34 authors' estimate (level 3).

Get more precise: requirements, red flags, horizon →

All details

If you want to know first **what drives this number**: open the map of the model → (level 2 — the click switches you there).

METHOD, LIMITS, DATA PROTECTION

Method: For each approach the calculator adds up licence and support (licensed headcount × price × years; licensed headcount = all report authors plus the share of viewers with access to standard reports), build-out (setup, optional in-house development, chart types × hours × blended hourly rate), rollout (reports × charts × reuse hours), training (report authors × hours and course fees, recurring through staff turnover), licence with viewer growth and price escalation per year, maintenance (rate × build-out + rollout × years plus breaking updates) and discontinuation risk (probability × migration share × build-out). Every assumption is a three-point estimate from which 1,500 scenarios are drawn via a PERT distribution; P10, P50 and P90 are shown. Break-even and sensitivity use the PERT expected value. Knock-out criteria (a must with 0 points, governance switches) filter before the ranking; excluded approaches stay visible for orientation. The requirements score is a weighted scoring (must 3, should 2, could 1 as weights, fulfilment level 1 to 5) and deliberately enters the ranking only through the slider. Governance and user support grow with the number of viewers (log-linearly from 50 to 5,000 viewers); the operating model separates deeply trained template builders from template users and scales variants, support and governance. Licences grow with the licensed headcount and the price, hourly rates with wage inflation, the running internal hours fall with the productivity gain, and maintenance follows the trend by cohort age. The

discontinuation risk is a hazard rate per year: in the expected value the sum over the years, in the simulation a drawn event year, with the damage always measured against that year's estate. The net present value applies the discount rate to the cash flow of each year. The ranking uses the cost ratio to the cheapest option (excluded ones included), so that a hopeless third option does not shift the order of the others; ranks with overlapping P10–P90 bands are merged. Amounts are rounded to two significant figures. Audit 09/2026 (three Opus and three Sonnet reviews, blind estimate, web verification): turnover training differentiated by role, discontinuation risk including 60% of the rollout, approval per visual package instead of per chart type, licence administration degressive and zero for a site licence, tiering per year, governance log-linear, valuation of internal hours as a constant; 22 defaults reset, provenance per value in the badge. **Every change since v0.8 is listed in the change history below.**

Change history v0.8 to today (15 entries)

- **v0.8** — chart mix by complexity (factors per approach, the panel hours apply to 50 / 35 / 15), Wright learning curve on reuse (panel value = the 10th chart per report author, floor 50%), the do-nothing option "no standard" as a benchmark outside the ranking, with reader time reported separately.
- **v0.10** — susceptibility to variants as an assumption of its own per approach instead of being tied to the design score, a "template library available" switch for Deneb and core (off by default, a project in its own right), price anchors on level 1 showing who leads under all three anchors. v0.11 (assumption audit round 2, package A): paired draws — one random stream per parameter, shared by all four approaches, switchable; rollout carries wage inflation on the growth arising over the horizon; cash out per year split into cash outflow and internal time (table and Excel sheet); payback against the do-nothing option with and without reader time; internal hours and FTE per approach with a capacity warning (1,500 h per FTE and year); decision measure P50 / P75 / P90 switchable (ranking by P90 = conservative); exit cost as a separate figure outside the total; a viewer cap against unlimited growth; the do-nothing option using the core complexity factors of the chart mix; the 50% floor applied to the product of library × learning curve instead of to the learning curve alone; new defaults with sources for hourly rates, wage inflation, licence price escalation, the Zebra anchor (derived), site licence, discontinuation risk for core and running governance for Deneb (2.0 on 8 September 2026).
- **v0.11 (package B)** — the red flag "certified visuals only" made precise (the tenant setting applies to AppSource, not to the Organizational Store) and a modifier "open-source visual via the Organizational Store" added, which lifts the knock-out without reducing the approval hours; three new requirements (AI-assisted authoring via PBIR / agent skills, accessibility, data export from the visual) and a corrected criterion "large data volumes", because the 30,000-row window is the SDK limit of every custom visual; source and note texts put right (reader-time study blueforte / TU München 2019 instead of Zebra BI, a currency band instead of "$1 ≈ €1, conservatively", a price increase as a change of model, the free desktop routes, cx0 and the learning-curve anchor named).
- **v0.12 (package C1, calculation core)** — the core now calculates year by year instead of with products of averages — estate and rollout cohorts per year, maintenance only on the existing estate and ageing from its own year of creation, licence with tiering, minimum bundles and a viewer cap per year, wage inflation and productivity gain per year, the learning curve on the cumulative number of charts built per report author. This makes the net present value, the cash flow and the cumulative chart dependable for the first time (the nominal total changes by only −4.5 to +0.1%, because a rollout that was too cheap and maintenance that was too expensive used to cancel each other out). Also new: a rollout period for the existing estate (no longer everything in year 1), a productivity gain per approach as the counterpart to wage inflation, a maintenance trend by cohort age instead of decay by calendar year, discontinuation as a hazard rate with a drawn event year, a shared team factor per scenario across all four approaches (the level band gets wider, the difference band stays narrow), minimum bundles and a minimum contract term for paid, lock-in on exit per approach, aligned complexity factors, and the marginal effort of the final year instead of the average over the horizon for "each further report". v0.12 (package C2, requirements and outputs): seven new requirements (control over the visual version, travelling with Git and a deployment pipeline, theme controllability, drilldown/drillthrough/report-page tooltip as a row of its own beside interactivity, which is now narrowed to selection and cross-highlighting, mobile, copying to PowerPoint, paginated reports) — 27 criteria in all; six new red flags (German public procurement, co-determination under § 87 Abs. 1 Nr. 6 BetrVG and the internal audit's right to inspect the calculation logic as warnings; no full input VAT deduction, budget approval only in the following year, and capacity at its limit as modifiers with a cost effect in the calculation core); four new inputs (adoption rate per approach, with the non-compliant remainder billed at do-nothing-option rates, semantic models as a multiplier on setup and initial build, distribution channel AppSource / Organizational Store / file import with factors on approval, running governance, breaking updates and lead time, and the lead time to production release as a display without cost effect); four new outputs (capex/opex split with amortisation and P&L impact under § 248 Abs. 2 HGB, IAS 38 and § 5 Abs. 2 EStG; an hours

anchor panel / blind estimate / conservative + 50% for the four unmeasured hours figures; switchable uncertainty in the volume assumptions of ±10 or ±20%, paired across all approaches; an evidence bar that splits the P50 cost per approach by the provenance of the driving assumptions into evidenced, rule of thumb and estimate). Updated: Fabric SKU presets F2/F8/F64. Details in the Markdown version. The Monte Carlo page (step 6) additionally draws the volume assumptions per scenario with the chosen variation band and evaluates the distribution, the certainty of the ranking, cash against internal time, building against using, rank correlations (Spearman) and a scale simulation at a fixed report authors : viewers ratio. Paginated reports (RDL) are not included: Power BI custom visuals do not exist there, neither purchased nor Deneb nor core — a printed report pack is built a second time under all four approaches. Power BI base licences (Pro, PPU, capacity base price) fall due equally for all approaches and are not included, nor is the benefit side (only as a score and as "€ per requirement point") or tax questions (e.g. § 50a EStG for licences from foreign vendors). Capitalisation and amortisation have been an output of their own since v0.12 — as an orientation, not as accounting or tax advice.

- **v0.13** — decision path, impact map, decision tree.
- **v0.14** — report classes — the reports are split into three classes (A operational with no IBCS need, B IBCS standard, C IBCS demanding), each with its own share, its own number of charts per report and its own chart mix. Charts per report and the global chart mix have been derived figures ever since; an expert override can still force a single global mix. Class A reports are built with core visuals and therefore cost the core hours in every approach (a proportionate share of the initial build of the chart types, rollout, maintenance, migration); they still drive licence, governance and user support, but not the approach-specific chart hours. The switch "IBCS mandatory for all reports" pulls class A into the chosen approach as well and makes the conformance review compulsory for every report; without it, only what belongs to classes B and C is reviewed. The review is a new global assumption (hours per report and review, plus an annual sample from the estate), the same for all four approaches, because checking a report against the IBCS rules is a human pass and does not depend on what the chart was built with — both values are authors' estimates with no source. The do-nothing option knows the classes too (with no standard every report is produced ad hoc, and the class mixes now only set the complexity) and has no conformance review: with no standard there is nothing to review.
- **v0.15** — vendor quotes — up to three real quotes can be entered in level 2 (vendor, model per user / volume tiering / site flat fee, price, number of users, minimum term, discount, support, price escalation clause, note). Exactly one can be active; for paid it replaces the price anchor, sets the licence model, the price or site flat fee, the minimum contract term, the price escalation clause and support as point values, and switches the volume tiering off, because a quoted price has already been negotiated (tiering on top would be a second discount). No new calculation routes arise from it. From two quotes onwards a table compares total cost, cost per viewer and year, rank among the four approaches, and the negotiation ceiling against the cheapest hours-based approach. A quote is the hardest evidence in the model and gets a source badge of its own (A); in the evidence bar the licence price has since counted as evidenced only when a quote is active — the three price anchors are search snippets and, without a quote, run in the middle class, which lowers the evidenced share of paid without a quote and lifts it markedly with one.
- **v0.16** — guided start — a mode ahead of level 1 (seven questions: report authors and viewers or a preset, the number of standard reports and the share of reports needing IBCS via three report-class templates, operating model, five common red flags, price anchor or vendor quote, horizon, close) that only sets existing fields and clicks the existing controls; no new calculation logic, no changed assumptions, no interim state of its own — every answer lands in the inputs at once and the result recalculates live. Alongside it a reading guide in the result section of level 2 that jumps in four steps to the decision path, the ranking with its bands, the report classes and the evidence bar, plus simplifications on levels 1 and 2: technical terms (P50, PERT, MoSCoW, knock-out, volume tiering, net present value, red flag, price anchor) are explained in half a sentence where they first appear, the glossary "Key terms explained" has moved forward from level 3 to level 2, charts per report are rounded to one decimal place in running text, and four fine adjustments (semantic models, adoption rate, uncertainty in the volume assumptions, hours anchor) have moved from level 2 to level 3 — with a note at their old place, unchanged defaults and an unchanged calculation. A link with settings skips the guided start; the guided start itself is not stored in the link (sv stays 4) and does not appear in the printout.
- **v0.17** — table of contents as a sidebar that travels with you from level 2 onwards (all sections and the main blocks of the result, grouped by page area (by level up to v0.18), the active position tracked by IntersectionObserver; entries on deeper levels switch the level when clicked instead of greying out, and open closed details) with the buttons "Expand all" and "Collapse all" for reading in one go and for printing; a clickable impact map — selecting an input, an assumption group or a cost item highlights the connected paths, fades the rest back and explains underneath where the selection feeds in and how much it accounts for (amounts and shares from the same categories as before, provenance of the assumptions as in the evidence bar); the sensitivity line "all assumptions in this group 10% higher ≈ €x" uses expected(key, override), i.e. the route of the tornado chart, and is

deliberately not shown for volume-assumption inputs, because there is no clean way to do it for them. On top of that the map becomes a second user interface: clicking a node opens a popup docked to that element — for inputs the matching existing control (number field, segment switch, report classes), for assumption groups the assumptions of that group with min · likely · max, source badge and short rationale as on level 3, for cost items only the composition; derived figures (charts per report, chart mix) stay unchangeable and point to the report classes. Every popup shows the effect as the result before and now, for the selected approach and for the cheapest permissible one, and has a "reset" button back to the state it had when it was opened. Only the existing fields and handlers are operated; no new calculation logic, no new state and no new share-link entry arise (sv stays 4). The map can also be reached directly from the table of contents, from the "Model as a map" button in the toolbar and from a third option at the end of the guided start.

- **v0.18** — licensed headcount separated from the volume assumptions — an independent review (causal-relationship check 09/2026, finding 1) established that although the hours of the report classes have been calculated separately since v0.14, the licence volume for 3rd-party paid still ran across all viewers, even though some of the readers never open a report built with the purchased visual. The new global assumption for this is "share of viewers with access to standard reports" (60 · 85 · 100%): only that share of the viewers plus all report authors make up the licensed headcount, and volume tiering and the minimum bundle apply to it as well; governance, user support, access governance and reader time stay on the full number of viewers, and the site licence is untouched. With the "IBCS mandatory for all reports" switch the share is necessarily 100% — the mandate lifts the licensed headcount back to all readers, and that is the second, so far invisible half of its price. The mechanism is evidenced by a search snippet (Zebra BI and Inforiver tie the licence scope to access to the reports built with the tool, not to the tenant); the level is an authors' estimate. **The correction works in favour of 3rd-party paid and against the approaches the authors run themselves**: the licence block of paid, and with it the tipping point, shift by roughly the size of the viewer share taken out; the negotiation ceiling has since been labelled as a price per **licensed** user and year and sits correspondingly higher. Links from older versions (sv 4 and earlier) keep calculating unchanged, because they are set to 100% when loaded.
- **v0.19** — information architecture after Shneiderman — overview first, then details. Up to v0.18 the impact map was a collapsed details block on level 3 at the foot of the page; it now stands as an open section of its own, "What drives the cost", on level 2 directly after the overview and before the inputs, because it is the overview of the model and not one of its details. The detailed reading instructions sit below it in a collapsed block; the decision tree without a scenario of its own stays a detail of level 3. Since then the table of contents groups by page area (guided start, overview and map, inputs, result, details) instead of by level, because the order of the page and the order of the levels are not the same. Calculation logic, assumptions and share link (sv stays 5) are unchanged.
- **v0.20** — Accessibility, states and keyboard; compliance check in the cash flow, VAT on course costs; print as a conference handout.
- **v0.21** — Debounced recalculation of the assumption fields; controls scoped to their own container (freeze in the map popup fixed).
- **v0.22** — One file, two languages: DE/EN switch top right, dictionary inside the document; expert-panel origin in the evidence bar and legend; long jumps snap instead of gliding for seconds.
- **v0.23** — Review round 3: tipping point and negotiation ceiling each come from one function, solved on the model instead of approximated (before: two values on one page); volume tiering as the cheapest covering tier, so Paid never gets cheaper with more viewers (where the next tier with its minimum quantity is cheaper, the Paid licence drops: enterprise preset −7%); the decision tree computes its ratios live; upper bounds for learning rate, external share and productivity gain; budget year fixed; open-source assumptions checked against manifest and docs (data processing and accessibility 3 each, texts and sources); five scenario presets. Share link sv stays 5.

All figures are indications under uncertainty, not quotes — with one exception: the figures you entered yourselves from a vendor quote. The calculation runs locally in the browser. The share link carries your inputs unencrypted in the URL fragment; whoever has the link sees the assumptions — **including the vendor quotes you entered, with vendor, price, terms and note, which also appear in the Excel sheet "Quotes"**. Fonts and the Excel library SheetJS are served by this website itself, not by third parties

Markdown version (model, assumptions, sources) · Visual Standards Calculator v0.23 · Michael Tenner & Diana Ackermann · Daten-WG Knowledge Kitchen · as of 11/2026 · Pre-Conference "Visual Standards in Power BI für Controlling und Finance"

Daten-WG · *Knowledge Kitchen*

Podcast, Tutorials, Werkzeuge und Datenstorys rund um Power BI, Microsoft Fabric und ehrliches Reporting.

FOLGEN

- Alle Folgen
- Start hier
- Suche
- Library

LERNEN

- Dein erstes Dashboard
- Power BI von A bis Z
- Microsoft Fabric End-to-End
- IBCS-Trainer
- Fabric Alchemie
- Daten-Imperium

WERKZEUGE

- MockupKitchen
- Chart-Builder
- ChartKitchen Visual
- P&L-Visual
- **Visual-Standards-Rechner**
- Alle Werkzeuge

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